

SAI CHAITANYA REDDY POREDDY | Data Engineer

saichaitanyaporeddy@gmail.com | +1 908-251-8454 | Bridgeport, CT | www.linkedin.com/in/chaitanyareddy399

SUMMARY

- Over 5 Years of IT experience and Currently working in a Big Data Capacity with the help of Hadoop Eco System across internal and cloud-based platforms.
- Excellent Working knowledge of Hadoop, Hive, Sqoop, pig, HBase & Oozie in real time environment and worked on many modules for performance improvements and architecture designing.
- Experience in Machine Learning, Data Mining with large datasets of Structured and Unstructured data, Data Acquisition, Data Validation, Predictive modeling.
- Experience in implementing various Big Data Engineering, Cloud Data engineering, Data Warehouse, Data Mart, Data Visualization, Reporting, Data Quality, and Data virtualization Solution
- Experience in Data transformation, Data mapping from source to target database schema, Data Cleansing procedures
- Adept in programming languages like Python including Big Data technologies like Hadoop, Hive
- Experienced in Normalization (1NF, 2NF, 3NF and BCNF) and De-normalization techniques for effective and optimum performance in OLTP and OLAP environments
- Excellent Knowledge of Relational Database Design, Data Warehouse/OLAP concepts, and methodologies
- Experience in designing star schema, Snowflake schema for Data Warehouse, ODS architecture
- Expertise in OLTP/OLAP System Study, Analysis and E-R modeling, developing Database Schemas like Star schema and Snowflake schema used in relational, dimensional, and multidimensional modeling
- Experience in designing, building, and implementing complete Hadoop ecosystem comprising of MapReduce, HDFS, Hive, Impala, Pig, Sqoop, Oozie, HBase, MongoDB, and Spark
- Experience in coding SQL for developing Procedures, Triggers, and Packages
- Experience writing spark streaming and spark batch jobs, using spark MLLib for analytics
- Defined real time data streaming solutions across the cluster using Spark Streaming, Apache Storm, Kafka, Nifi and Flume
- Strong experience and knowledge of real time data streaming analytics using Spark Streaming, Kafka, and Flume.
- Having experience in developing a data pipeline using Kafka to store data into HDFS
- Strong work experience on Kafka streaming to fetch the data real time or near real time
- Experienced in Spark Core, Spark RDD, Pair RDD, Spark Deployment Architectures
- Involved in queue extraction Dstreams from near real-time data using Kafka & Spark Streaming with Scala and also have development experience using Rest with EMR, Spark- Core, Context, DataFrame API, RDD and Spark SQL with build tools such as Maven & Sbt

TECHNICAL SKILLS

Methodologies: Agile (Scrum, Kanban), Waterfall

Database: MySQL, Teradata, Oracle, Sybase, SQL server, DB2

Cloud: Snowflake, AWS Redshift, S3, Redshift Spectrum, RDS, Glue, Athena, Lambda, Cloudwatch, HIVE, Presto.

Data Orchestration Tools: AWS Managed Apache Airflow(MWAA) & Splunk(logs), Snaplogic.

DevOps: Docker Images, Kubernetes Container, CI/CD Pipeline

ETL Tools: Informatica Power Center 8.x,9.x,10.x, Informatica Intelligent Cloud Services

Data Virtualization tools: Denodo, Power BI

Scheduling: Active Batch, CA7, Control- Confidential

Scripts: Advanced SQL, Python, PL/SQL, Unix shell scripting, XML, JSON

Code Mgmnt process/tools: GitLab, GitHub, Bitbucket, Microsoft TFS, SourceTree

Project Mgmnt Process: JIRA, Confluence

Defect Tracking tools: HP Quality Center, Microsoft TFS

Design: Microsoft Visio.

Operating System: Windows, Linux

PROFESSIONAL EXPERIENCE

Data Engineer/ Developer

Citi Bank – New Jersey

June 2024 – Present

- Worked as Data Engineer to review business requirement and compose source to target data mapping documents
- Involved in Agile development methodology, active member in scrum meetings
- Involved in Data Profiling and merge data from multiple data sources
- Involved in Big data requirement analysis, develop and design solutions for ETL and Business Intelligence platforms
- Designed 3NF data models for ODS, OLTP systems and dimensional data models using Star and Snowflake Schemas
- Worked on Snowflake environment to remove redundancy and load real time data from various data sources into HDFS using Kafka
- Developed data warehouse model in Snowflake for over 100 datasets
- Designing and implementing a fully operational production-grade large-scale data solution on Snowflake Data Warehouse
- Work with structured/semi-structured data ingestion and processing on AWS using S3, Python. Migrate on-premises big data workloads to AWS
- Designed the data aggregations on Hive for ETL processing on Amazon EMR to process data as per business requirement
- Involved in migration of data from existing RDBMS to Hadoop using Sqoop for processing data, evaluate performance of various algorithms/models/strategies based on real-world data sets
- Implemented Data Validation using MapReduce programs to remove unnecessary records before move data into Hive tables
- Created Hive tables for loading and analyzing data and developed Hive queries to process data and generate data cubes for visualizing
- Extracted data from HDFS using Hive, Presto and performed data analysis using Spark with Scala, PySpark and feature selection and created nonparametric models in Spark
- Handled importing data from various data sources, performed transformations using Hive, MapReduce, and loaded data into HDFS
- Worked on Enterprise Messaging Bus with Kafka-Tibco connector and published Queues were abstracted using Spark Streams and parsed XML, JSON data in Hive.
- Designed and configured Kafka cluster to accommodate heavy throughput of 1 million messages per second. Used Kafka version 0.6.3 API's to produce messages

Environment: Python, SQL, CVS/XML Files, Oracle, JSON, Cassandra, MongoDB, AWS, Azure, DataBricks, Snowflake, Hadoop, Hive, MapReduce, Scala, Spark, J2EE, Agile, Apache Avro, Apache Maven, AirFlow, Kafka, MLib, regression, Docker, Tableau, Git, Jenkins.

Big Data Developer

State Street — New Jersey

February 2023 – May 2024

- As a Data Engineer, provided technical expertise and aptitude to Hadoop technologies as they relate to the development of analytics.
- Responsible for the planning and execution of big data analytics, predictive analytics, and machine learning initiatives
- Very good hands-on experience in advanced Big-Data technologies like Spark Ecosystem (Spark SQL, MLib, SparkR and Spark Streaming), Kafka and Predictive analytics (MLib, R ML packages including ML library of

H2O)

- Designed and developed spark jobs for performing ETL on large volumes of medical membership and claims data
- Created Airflow Scheduling scripts in Python
- Involved in importing the real-time data to Hadoop using Kafka and implemented the Oozie job for daily
- Developed applications of Machine Learning, Statistical Analysis, and Data Visualizations with challenging data processing problems
- Compiled data from various sources public and private databases to perform complex analysis and data manipulation for actionable results.
- Designed and developed Natural Language Processing models for sentiment analysis.
- Worked on Natural Language Processing with NLTK module of python for application development for automated customer response.
- Demonstrated experience in design and implementation of Statistical models, Predictive models, enterprise data model, metadata solution and data lifecycle management in both RDBMS, Big Data environments.
- Used predictive modeling with tools in SAS, SPSS, and Python.
- Applied concepts of probability, distribution, and statistical inference on given dataset to unearth interesting findings through these of comparison, T-test, F-test, R-squared, P-value etc.
- Applied linear regression, multiple regression, ordinary least square method, mean-variance, the theory of large numbers, logistic regression, dummy variable, residuals, Poisson distribution, Bayes, Naive Bayes, fitting function etc. to data with help of Scikit, SciPy, NumPy and Pandas module of Python.
- Applied clustering algorithms i.e. Hierarchical, K-means with help of Scikit and SciPy.
- Developed visualizations and dashboards using ggplot2, Tableau
- Worked on development of data warehouse, Data Lake and ETL systems using relational and non-relational tools like SQL, No SQL.
- Built and analyzed datasets using R, SAS, MATLAB, and Python (in decreasing order of usage).
- Used Agile Methodology for the software development life cycle.

Environment: Spark, AirFlow, Machine learning, AWS, MS Azure, Cassandra, Avro, HDFS, GitHub, Hive, Pig, Linux, Python (Scikit-Learn/SciPy/NumPy/Pandas), SAS, SPSS, MySQL, Bitbucket, Eclipse, XML, PL/SQL, SQL connector, JSON, Tableau, Jenkins.

Associate Data Engineer

CloudEnd Platform Pvt Ltd. — Hyderabad, India

May 2021 – July 2022

- Worked with Hadoop ecosystem covering HDFS, HBase, YARN and MapReduce.
- Worked with Oozie Workflow Engine in running workflow jobs with actions that run Hadoop MapReduce, Hive, Spark jobs.
- Performed Data Mapping, Data design (Data Modeling) to integrate data across multiple databases into EDW.
- Responsible for design and development of advanced Python programs to prepare, transform and harmonize data sets in preparation for modeling.
- Hands-on experience on Hadoop/Big Data related technology experience in Storage, Querying, Processing, and analysis of data.
- Developed Spark/Scala, Python for regular expression (RegEx) project in Hadoop/Hive environment for big data resources.
- Automated the monthly data validation process to validate the data for nulls and duplicates and created reports and metrics to share it with business teams.
- Used clustering techniques like K-means to identify outliers and to classify unlabeled data.

- Data gathering, data cleaning and data wrangling performed using Python.
- Transformed raw data into actionable insights by incorporating various statistical techniques, data mining, data cleaning, data quality, integrity utilizing Python (Scikit-Learn, NumPy, Pandas, and Matplotlib) and SQL.
- Calculated errors using various machine learning algorithms such as Linear Regression, Ridge Regression, Lasso Regression, Elastic net regression, KNN, Decision Tree Regressor, SVM, Bagging Decision Trees, Random Forest, AdaBoost, and XGBoost. Chose best model eventually based on MAE.
- Experimented with Ensemble methods to increase accuracy of training model with different Bagging and Boosting methods.
- Identified target groups by conducting Segmentation analysis using Clustering techniques like K-means.
- Conducted model optimization and comparison using stepwise function based on AIC value.
- Used cross-validation to test models with different batches of data to optimize models and prevent overfitting.
- Worked and collaborated with various business teams (operations, commercial, innovation, HR, logistics, safety, environmental, accounting) to analyze and understand changes in key financial metrics and provide ad-hoc analysis that can be leveraged to build long term points of view where value can be captured.
- Created and analyzed customer specific features by using Matplotlib, Seaborn in Python and dashboards in Tableau.
- Gained domain knowledge and application portfolio knowledge to play a key role in defining the future state of large, business technology programs.
- Used Kibana, an open-source plugin for Elasticsearch in analytics and Data visualization.
- Developed MapReduce/Spark Python modules for map-reduce and predictive analytics in Hadoop on AWS.

Environment: Erwin, Python, SQL, SQL Server, Informatica, SSRS, PL/SQL, T-SQL, Tableau, MLlib, Spark, Kafka, MongoDB, logistic regression, Hadoop, Hive, OLAP, Azure, MariaDB, SAP CRM, HDFS, SVM, JSON, Tableau, XML, AWS.

EDUCATION

Masters in Computer Science

University of Bridgeport, Bridgeport, CT | Aug 2022 – May 2024

Bachelor of Technology in Computer Science & Engineering

Lovely Professional University, Punjab, India | Jun 2017 – Aug 2021

CERTIFICATIONS

- **Microsoft Certified: Azure Data Engineer Associate**
- **AWS Certified Solutions Architect**
- **Certified Scrum Master (CSM)**